

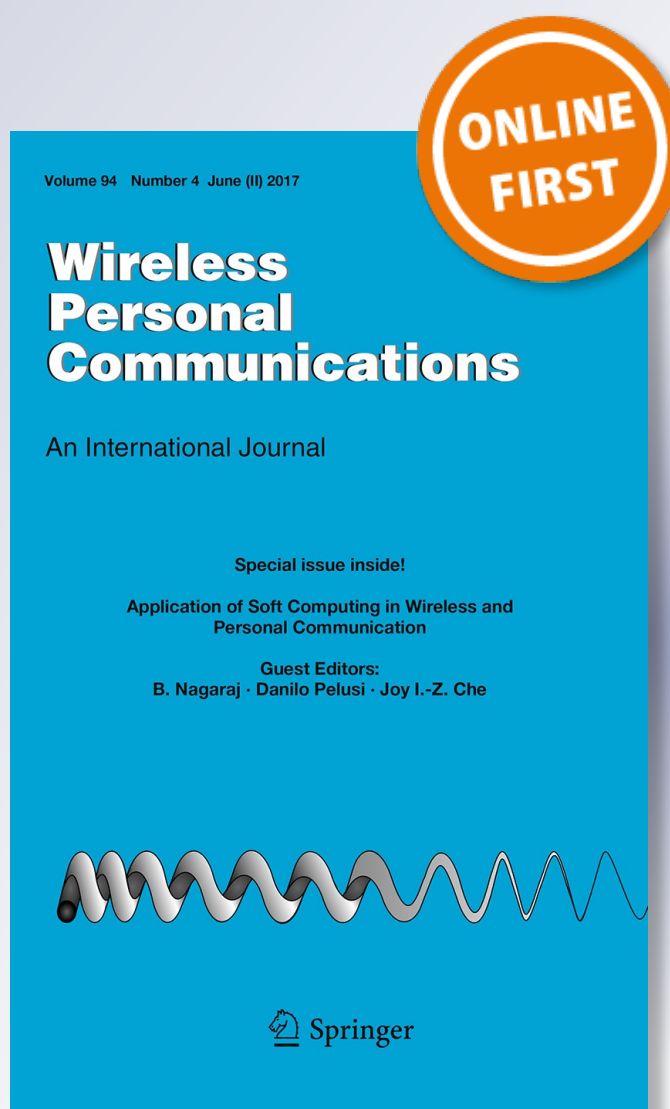
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Cyber Security Threats to IoT Applications and Service Domains

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Abstract We are currently living in the post-PC era where smartphones and other wireless handheld devices are changing our environment, making it more interactive, adaptive and informative. Termed as Internet of Things (IoT) evolving into Internet of Everything, the new ecosystem combines wireless sensor networks, cloud computing, analytical data, interactive technologies, as well as smart devices, to provision solutions in which the objects are embedded with network connectivity and an identifier to enhance object-to-object interactions. IoT innovation is advancing and provides diverse smart solutions or applications. From e-transport to e-health; smart living to e-manufacturing and many other e-solutions. In this environment, the rising trend of cyber attacks on systems infrastructure coupled with the system inherent vulnerabilities presents a source of concern not only to the vendors, but also to the consumer. These security concerns need to be addressed in order to ensure user confidence so as to promote wide acceptance and reap the potentials of IoT. From the perspectives of firmware, hardware and software infrastructure setups, this paper looks at some of the major IoT application and service domains, and analyze the cybersecurity challenges which are likely to drive IoT research in the near future.

Keywords Internet of Things · Internet of Everything · Cyber threats · Systems vulnerabilities · IoT infrastructure · IoT applications

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1 Introduction

According to IBM, devices connected to the Internet are expected to exceed the number of human beings and the evolution of connectivity is expected to continue such that by 2020 the number of connected devices will be around 50 billion [1]. This proliferation of connected devices in an actuating network has created what has become known as the Internet of Things (IoT). A platform in which sensors and actuators blend seamlessly with the environment to share information in order to develop a common operating picture [2]. An IoT system starts from the level where a single object is identified using a unique global identifier which is globally addressable. The level of information obtained by accessing the object, in this case, can be as low as static data that is stored on the radio frequency identification (RFID) tags. IoT is therefore described as objects with a unique identifier, having Internet connectivity; is (interactively) accessible by other objects herein referred to as the “things”. IoT has stepped out of its infancy and is the next revolutionary technology in the transformation of Internet into a fully integrated future Internet (of things). This development is fuelled by the recent increase in adoption and integration of wireless network technologies, Wireless Sensor Networks (WSN), RFID tags, as well as actuating nodes. On this concept, Karimi and Atkinson claimed, expanding communication networks to include physical objects will further accelerate the number of connected devices, as well as the amount of information that can be shared through the Internet [3].

IoT presents ubiquitous connectivity for a wide range of devices, services, and applications. These include intelligent computers, smartphones, office equipment, wireless-enabled cars, lighting systems, heating, and ventilation and air-condition (HVAC), household appliances, and many others. To be IoT-enabled, a device (‘thing’) ought to be on a network and connected to a communicating node. Various communication network technologies (infrastructures) such as 3G, LTE, Wi-Fi, Bluetooth, ZigBee, Z-wave, Sigfox, etc. provide connectivity services for IoT deployment on many services platforms.

As IoT advances, cloud computing is expected to provide the backbone for the worldwide information diffusion, data analytics (or computation) and storage. Cloud solutions such as Microsoft Azure, Amazon Web Services (AWS), Google Docs, etc., are expected to provide standard gateways for interconnecting physical objects with computation and communication capabilities across a wide range of applications, services, and technologies. Caceres and Friday [4] discuss the progress, opportunities and challenges of an environment of ubiquitous computing ‘ubicom’ and identify the two critical technologies for growing the future ubicom infrastructure as cloud computing and the Internet of Things. It is thus generally perceived, that as IoT matures, cloud computing will act as a receiver of data from the various ubiquitous sensors, as well as being the platform for big data analytics, analyzing and interpreting IoT generated data [4]. Additionally, various cloud-based solutions increasingly are provisioned to provide users with compatible web-enabled interface applications for user interactivity and connectivity.

For instance, Gubbi et al. [2], have argued that for IoT to emerge successfully, the traditional computing and Internet connectivity platforms will have to be extended beyond the traditional mobile communication connecting human beings, and evolve into connecting objects and embedding intelligence into our environment. With this fundamental ground in place, smart connectivity and context-aware computation can be accomplished. Certainly, with billions of devices expected to be connected to the IoT ecosystem, it is expected to generate enormous amounts of data which will have to be stored, processed and presented in a seamless, efficient, and easily interpretable form. In that case, cloud-

centric computing would be required as the platform to provide virtual infrastructure support for IoT services [2]. The services will integrate monitoring, storage, computation (and/or analytics), visualization and client service delivery [2]. According to Gubbi et al., the vision of IoT can be seen from the 'Internet' and 'Thing' centric. The Internet-centric architecture embroils Internet-as-a-services being the core focus of IoT while data is contributed by the objects [2]. In the thing-centric architecture, smart objects take the center stage of IoT services and applications [5].

The major concerns raised with IoT-Cloud integration is the fact that from infrastructural to service domains, cloud models are beset with various security challenges such as application services attack, data integrity attack, privacy, trust, identity, standardization, etc. These challenges are likely to unveil when the two platforms merge raising some fundamental research questions which need to be explored. Accordingly, a study by International Data Corporation (IDC) claims, that while business leaders recognize the business potentials of IoT, they are deeply skeptical about the system's inherent security challenges [6]. Additionally, the study asserts, most business leaders admitted having little understanding or underestimate the security threats IoT brings. In a related study, KPMG stated, that the security breaches in consumer data as well as recent attacks on cyber infrastructure systems worldwide, make IoT users lose confidence and avoid solution providers who fail to take the appropriate measures to protect their systems [7]. The issue of cybersecurity challenges on IoT platforms is a major global concern which requires a holistic assessment from both research and industrial communities. This paper assesses security related threats which are likely to impede the success of IoT adoption and consumer confidence.

The paper is structured as follows: Sect. 2 provides an overview of the IoT concept and structures. In Sect. 3, the potential threat challenges in relation to IoT applications and service are assessed. In Sect. 4 domain scenarios are outlined and the Smart Metering case presented. In Sect. 5 Cyber Threat in Smart Metering is analysed as a case. Section 6 summarizes the conclusions.

2 The IoT Concept

Over the past decade, Internet technologies have revolutionized the interconnection among people at an unprecedented scale and pace. The next revolution is expected to craft the interconnection among diverse objects leading to what experts termed as the smart environment. As we move from www (static pages web) to web2 (social networking web) to web3 (ubiquitous computing—or web of things), the need for data-on-demand using sophisticated intuitive queries continues to increase significantly [2]. This era could be termed as the post-PC era where smartphones and related devices are changing our environment and the way “things” (including humans) interact. Things in the new environment are becoming more interactive as well as informative. Mark Weiser (father of Ubiquitous Computing), defined the new ecosystem as the “smart environment in the physical world that is richly and invisibly interwoven with sensors, actuators, displays, and computational elements, embedded seamlessly in the everyday objects, and connected through a continuous network” [8]. As mentioned, Gubbi et al. [2] argued that the growth of ubiquitous computing are shaped by cloud computing and the IoT.

IoT as a concept is presumed to have been coined by Kevin Ashton in a submission, where he argued “adding RFID and other sensors to everyday objects will create an

Internet of Things, and lay the foundations of a new age of machine perception” [9]. Since then, the idea has advanced in its acceptance both in the research and industrial ecospheres. Roman et al. [10] argue that basically, an IoT device as a heterogeneous object will have a locatable, addressable, and readable counterpart on the Internet whereby IoT opens up a communication channel with any other entity, providing and receiving services at any time, any place, and in any way. In that perspective, most ‘things’ (e.g. Human beings, pets, farm animals, and computers, books, cars, household appliances and food) will be on the Internet in one form or the another leading to the evolution of Internet of Everything (IoE) (Fig. 1) exhibiting a plethora of features (Fig. 2).

In 2005, the International Telecommunication Union (ITU) proposed the concept of Internet of Things in the ITU Network Reports 2005 [11]. The report described the era of IoT as the era when the car will alarm automatically when the driver makes a mistake; when the briefcase could remind the carrier what he forgot to take; the clothes will tell the washing machine their colours and temperature requirement, etc.

The European Research Cluster on Internet of Things (IERC) defines IoT as an integrated part of the future Internet with the following characteristics (Fig. 3).

Since then, a number of studies have looked at the various aspect of the IoT. For example, Roman et al. [10] provided the explicit analysis of the features and security challenges of Internet of Things in the perspective of distributed systems. Among the issues discussed were identity and authentication, access control, protocol, fault tolerance, trust, and governance.

2.1 IoT Infrastructure

It is established, that generally the IoT infrastructure (Fig. 4) consists of diverse hardware resources such as WSNs, RFID tags, actuators, readers, cameras, controllers, GPS, sensors (magnetometers, waspmote, ultrasound and infrared), device processors, terminals and other sensor gateways). Moreover, at the firmware level, most IoT objects are embedded with silicon integrated circuits (IC) and nano-electronics focussing on miniaturization, low cost and increased functionality in design of wireless identifiable systems or communication-enabled nodes. RFID tags and WSN hardware remain two most prominent IoT hardware infrastructure resources. RFID technology enables data transmission over a short distance. It consists of either active or passive radio frequency (RF) tag attached to an item

Fig. 1 Futuristic Internet of Everything (IoE) ecosystem

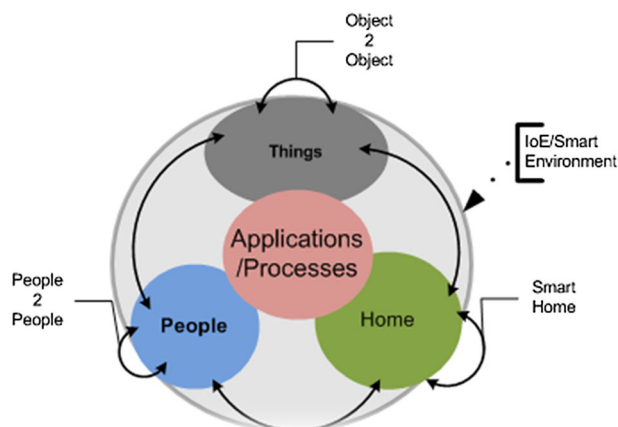


Fig. 2 Characteristics of futuristic IoE

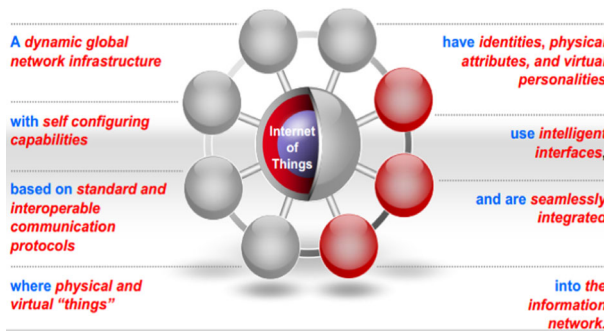
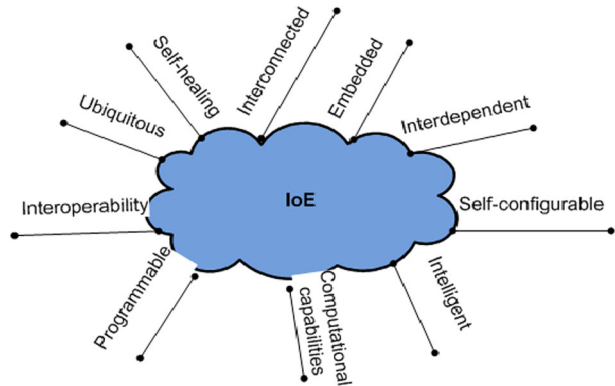


Fig. 3 IERC definition of IoT [22]

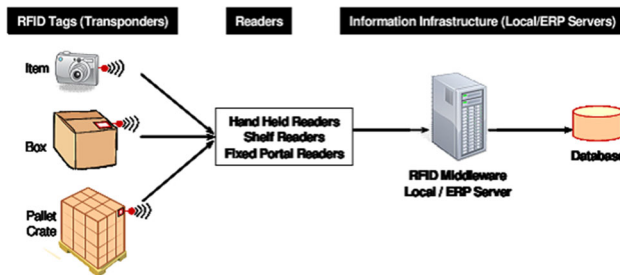


Fig. 4 RFID operational structure [23]

being tracked and an RF reader/emitter. A passive tag draws energy from its reader, whereas an active RFID tag draws power from its embedded device.

WSN hardware contains sensor interfaces, processing units, transceiver units and power supply.

IoT supporting software, on the other hand, varies from firmware operating systems to application software with cloud-based software solutions (SaaS) and Mobile OS (iOS, android, Blackberry) as the backbone. Other software solutions include network and device

OS which are mostly microkernel based (e.g. TinyOS, TinyDB, Nano-RK, LiteOS, VMs). Others include application development software such as HTML, JavaScript, Ajax, PHP, and Ruby. IoT software integration in addition to general software services also supports data visualization, system integration, remote access control and application programme interface (API).

3 IoT Cybersecurity Challenges

According to Roman et al. [10], one key challenge which must be overcome in order to push IoT into the real world is security. Security challenges relating to IoT line up with the traditional Information Systems (IS) security objectives (SO) which are confidentiality, integrity, and data availability [12]. Moreover, there are other security challenges which appear to be IoT-specific. For example, the merging of cloud computing and IoT, exposes IoT platforms to cloud induced vulnerabilities such as those contained in OWASP top 10 [13]. Figure 5 below presents cloud induced vulnerabilities identified in 2015 from the common vulnerability database [13]. These vulnerabilities which are inherent in cloud applications are likely to impact on IoT solutions and services as the two emerge.

Another significant risk vector may be found in substandard IoT products and services. These have the potential to threaten the survivability of IoT services. For instance, poorly designed, crafted and out-dated or counterfeit products present very significant risks to IoT enabled applications. On this subject, Touhill and Touhill [14] argue, businesses around the world suffer countless hours of monetary and mission loss due to unexpected equipment and system failures, caused by poor or improper maintenance, poor and inaccurate advice from an unqualified service personnel. Additionally, the authors claim poor performance by contracted personnel, and even inaccurate data from a business partner and perhaps IoT devices may be accepted for information processes and critical business decisions [14].

Furthermore, most IoT cybersecurity challenges lay in the system's own inherent vulnerabilities which expose the infrastructure setup to various attacks. The sources may include firmware, hardware (device), system applications, data, as well as the network interfaces or ports. Also, the bi-directional communication links between objects-to-objects leave the system open for network-related attacks and protocol failure. Other related

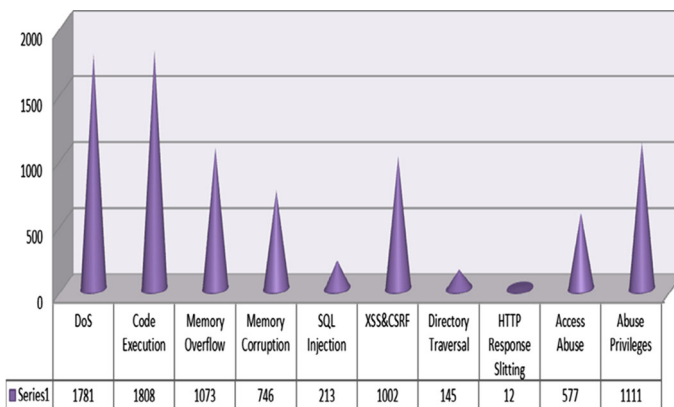


Fig. 5 CVE database, 2015 [24]

attacks include wireless scrambling, eavesdropping, man-in-the-middle attacks, and message modification and injection attacks. For example, IP-based devices are susceptible to IP misconfiguration which occasionally exhibits nondeterministic behaviour in terms of attack. IP misconfiguration inevitably decreases system performance and reliability. Moreover, while IoT and cloud computing integration scales up IoT applications and services, the integration also exposes IoT infrastructure and interdependent systems to public networks and global gateway. In addition to IP spoofing, service gateways (both local and global), can become perfect points for intrusions, DoS, injection attacks and other Internet/network-based attacks.

Added to these, most IoT user interactive applications are web and/or mobile based, designed mostly with application programming interface (API) as the interface code architecture using PHP, Java, XML, and HTML. An unpatched API may be susceptible to various attacks exposing the entire system to malicious attacks. For example, in CVE-2016-7413, the “use-after-free vulnerability in the wddx_stack_destroy function in ext/wddx/wddx.c in PHP before 5.6.26 and 7.x before 7.0.11 allows remote attackers to cause a denial of service via a wddxPacket XML document that lacks an end-tag for a recordset field element, leading to mishandling in a wddx_deserialize call” [13]. Similarly, in CVE-2016-4539, “the xml_parse_into_struct function in ext/XML/xml.c in PHP before 5.5.35, 5.6.x, 5.6.21, 7.x and before 7.0.6 allows remote attackers to cause a denial of service (buffer under-read and segmentation fault) or possibly have unspecified other impact via crafted XML data in the second argument, leading to a parser level of zero” [13].

In general, IoT systems are designed with security in mind, however, security misconfiguration can occur at any level of IoT communication architecture or any part of the system application. For instance, at the firmware level, IoT device such as RFID tags (chips) usually have internal memory, or self-powered batteries; power fluctuations in power grid or alternative currents (AC) could cause devices to lose memory leading to data loss. Furthermore, intermittent power fluctuations in semiconductor devices do result in signal loss which leads to potential system malfunction. Other security challenges in the IoT devices include component incompatibility, as well as device-based (physical) attacks, such as natural disasters, illegitimate use of the device (e.g. pilferage), and masquerading. To overcome these challenges will require innovative research and comprehensive system solutions focusing on system architectural design, firmware and hardware reconfiguration, network hardening and dynamic system application design.

3.1 IoT Cyber Attack

From the above challenges, we present a taxonomy of cybersecurity attacks on object-to-object communications by analysing system vulnerabilities vis-à-vis potential threat actors. In this taxonomy, six types of vulnerabilities are discussed. These are IP misconfiguration, injection, DoS, Code execution, Memory corruption, and XSS and CSRF. Corresponding threat vectors include physical (device) attack, application (software) attack, network attack, web interface attack, and data attack. Table 1 shows our proposed vulnerability–threat matrix. In columns 3 and 4, threat vectors are matched with their corresponding vulnerabilities.

Table 1 Vulnerability–threat matrix

Vulnerabilities (V)	Cyber-attack vectors (AV)	Vulnerability–threat matrix	
		Attack vectors	Vulnerabilities
IP misconfiguration (IM)	Device attack (DA)	DA	IP, MC, CE, D
SQL injection (SI)	Application attack (AA)	AA	SI, D, CE
DoS (D)	Network attack (NA)	NA	SI, D, CE
Code execution (DE)	Web interface attack (WiA)	WiA	SI, D, XC, IP
XSS and CSRF (XC)	Data integrity attack (DA)	DA	SI, CE
Memory corruption (MC)			

3.2 Attack Vectors

3.2.1 Device Attack

This attack type is capable of compromising IoT devices. Its main objective is to compromise critical architectural functions of the system (depending on the devices involved). In a RFID powered inventory control system, a successful attacker could bring the entire network down (especially when the base device (e.g. server is the target). In a neighbourhood area network (NAN) of a power grid, a device attack could affect the resistance of the network which in the extreme case may lead to distributed denial of service attacks on the entire grid. Device attacks may be caused by IP misconfiguration, memory corruption, and wrongly executed code in the device operating system at the middleware layer.

3.2.2 Application Service Attack

This is a type of attack that compromises system applications (Web, Mobile, System, etc.) which are run on various components of the system. A typical IoT application runs multiple sessions both at the local and the server levels. In most cases, the applications may be owned by application service providers (ASPs) or third party vendors. As indicated earlier, cyber attacks on these applications could surely compromise other interdependent systems. Common vulnerabilities in this type of attack include SQL injection, code execution, and DoS. Most IoT-enabled applications are designed using Raspberry Pi which is based on Linux kernel. Vulnerabilities in the kernel expose dependent applications to various forms of application attacks. For instance, in CVE-2016-8658, a “stack-based buffer overflow in the `brcmf_cfg80211_start_ap` function in `drivers/net/wireless/broadcom/brcm80211/brcmfmac/cfg80211.c` in the Linux kernel before 4.7.5 allows local users to cause a denial of service (system crash) via a long SSID Information Element in a command to a Netlink socket” [15]. Similarly, in CVE-2016-5344, “multiple integer overflows in the MDSS driver for the Linux kernel 3.x, as used in Qualcomm Innovation Center (QuIC) Android contributions for MSM devices and other products, allow attackers to cause a denial of service via a large size value, related to `mdss_compat_utils.c`, `mdss_fb.c`, and `mdss_rotator.c`” [15].

3.2.3 Network Attack

This is an attack which aims at compromising intercommunication among devices by either delaying message forwarding or message loss. Network attacks may destruct computational processes within the IoT configuration systems. In a home area networks (HANs), this type of attack aims at destructing the functionalities of the monitoring or interconnected devices. Similarly, in a neighbourhood area network (NANs), this type of attack could isolate or deny connected devices from accessing vital information from the neighbouring devices or addressing messaging request from neighbouring device. Causes of such attacks include SQL injection, DoS and code execution.

3.2.4 Web Interface Attack

This type of attack presents itself as a result of account enumeration, lack of account lockout or weak account credentials. In this case, an attacker uses weak authentication protocols (either capturing plain-text credentials or enumerating accounts) to accessing web interface. Web interface attacks may be caused by cross-site scripting (XSS), cross-site reference forgery (CSRF), IP misconfiguration and SQL injection. Other sources include insecure web interface design and weak account credentials. The attack compromises device integrity and could lead to denial of services. For instance in CVE-2016-7571, a cross-site scripting (XSS) vulnerability in Drupal 8.x before 8.1.10 allows remote attackers to inject arbitrary web script or HTML via vectors involving an HTTP exception [16]. Similarly, in CVE-2016-8581, a persistent XSS vulnerability exists in the User-Agent header of the login process of AlienVault OSSIM and USM before 5.3.2 allows an attacker to steal session IDs of logged in users when the current sessions were viewed by an administrator [17].

3.2.5 Data Integrity Attack

This is an attack whereby a threat agent attempts to compromise system data by inserting, altering or completely deleting data (either stored or in transmission) so as to deceive smart device to make wrong decisions or compromise its integrity. Data attacks may be caused by SQL injection and code execution which may be executed by a remote attacker.

4 IoT Application and Service Domains

Considering the recent advances made in IoT platforms, it is nearly impossible to envisage the numerous IoT applications, having in mind the continuous innovation in the technology, services and continuous needs in the industry. The current application domains include (but are not limited to) independent living (smart homes), smart cities, smart energy (smart grid and metering), smart mobility and transport, healthcare, retail and logistics, environmental monitoring, smart manufacturing (Fig. 6).

There are security challenges associated with all of these areas where the domain specifics raise different concerns. Some are very obvious and generally discussed as security breaches related to healthcare opening for misuse of very personal information; the etrade/retailing may open for monetary/financial misuse, but also the fast evolving

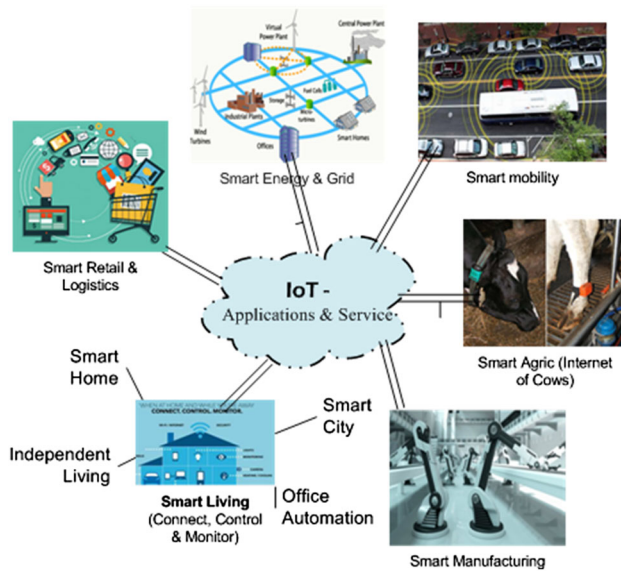


Fig. 6 Application areas of IoT

smart living/smart city domain raises security concerns related to the threats discussed in Sect. 3.

4.1 Smart Grid

There are huge possibilities to make a city ‘smart’ in the perspective of IoT. The domain covers activity sensing and events tracking, involving interactive objects in a smart environment. IoT application in a Smart City involves a huge variety of both infrastructure and technology requirements. One of the areas emerging with huge IoT potentials is the energy domain/smart energy.

Smart energy (grid) is a kind of “Internet” in which the energy packet is managed similarly to the data packet—across routers and gateways which can autonomously decide the best pathway for the packet to reach its destination with the best integrity levels [18]. The “Internet of Energy (IoEn)” concept is defined as a network infrastructure based on standard and interoperable communication transceivers, gateways and protocols allowing a real-time balance between a local and a global energy generation, storage, distribution and demand optimization. The IEEE 2030 standard on smart energy identifies Wide Area Network (WAN), Field/Neighbourhood Area Network (FAN/NAN) and Home Area Network (HAN) as the major components of a smart grid.

4.1.1 Smart Metering

According to Vermesan and Friess, IoEn is expected to provide an innovative concept for power distribution, energy storage, grid monitoring and communication as presented in [19]. IoT as a supporting platform supports energy distribution as when and where energy is needed. Thus, power consumption monitoring is performed at all levels; from local (home) devices to national and international distribution point [20]. The advanced

monitoring process in the power grid is known as smart metering [20] also known as advance metering infrastructure (AMI) (Fig. 7).

According to Azasoo and Tweneboah-Koduah, smart metering, has emerged as an innovative system for monitoring energy distribution, consumption, and billing via the convergence of power engineering, information and communication technology [20]. AMI provides a two-way information transfer for tracking of energy consumption (as a source of data for consumer billing) as well as tracking outages. Data gathered by the AMI is used to inform consumers of their energy consumption. AMI, in this case, provides a platform to interconnect smart meters, smart appliances, control devices, access points and the utility processing centers. In the smart energy scenario, smart meters give information about the instantaneous energy consumption to the user, thus allowing for identification and elimination of energy wasting devices and providing a platform for optimizing individual energy consumption as well as consumer billing.

5 Experimental Evaluation of Cyberattacks Against IoT Connected Device (SQLi and DoS Attack): Smart Metering Case

In this section, we demonstrate how SQL injection and DoS attacks are executed against IoT connected devices with smart metering as a case. The demonstration was performed on a self-configured virtual server. In both cases, the results show a successful cyberattack on connected IoT devices having a remote server (see Fig. 8 below).

SQL injection attack—Algorithm (Fig. 9)

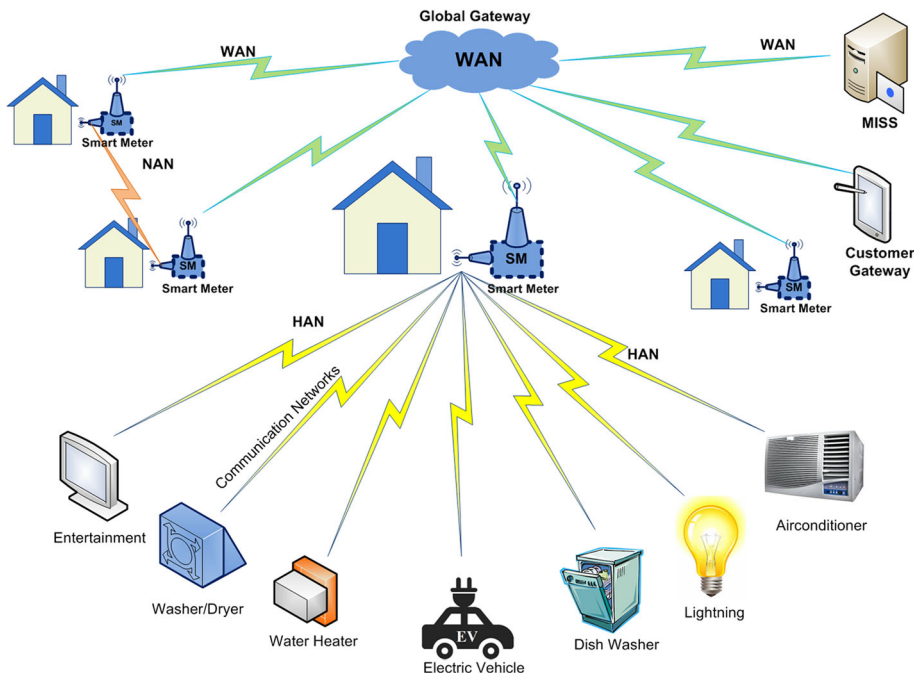


Fig. 7 Smart metering communication architecture [20]

```

Print header information
for URL in target URLs
    for payload in get request payloads
        response = send get request probe to server
        if response.status code == 500
            print payload and exist for manual attack
    for payload in post request payloads
        response = send post request probe to server
    
```

SQL injection attack (using Python)

This function delivers a payload to the smart metering gateway using an http 'get' method.

To do this, the payload is added to the url. The url sends a request to @params payload {string}. The request parameters for example `requests.get('http://www.test.com/', params=payload)` will map to `http://www.test.com/?key=value`

```

###
def http_get(url, payload):
    r = requests.get(url, params=payload)
    return process_responds(r)
###
    
```

The function processes a request to a smart metering gateway to determine if a probe is positive or negative. A positive probe indicates an open/unprotected port (vulnerability).

Probing Get (assuming query is contracted: where id = <defined_param>
 ('Params ', {'make': ""})
 ('Url: ',
 'http://metering.grid.com/metering/meter/topup_histo
 ry')

data been sanitised
 data been sanitised
 data been sanitised
 data been sanitised

Probbing Post

data been sanitised
 data been sanitised
 data been sanitised
 data been sanitised

Vulnerability: Weakness found (SQL injection)

Threat: data sanitised

Effect: sensitive information could be disclosed by injection attack

Impact: Data confidentiality and integrity could be compromised

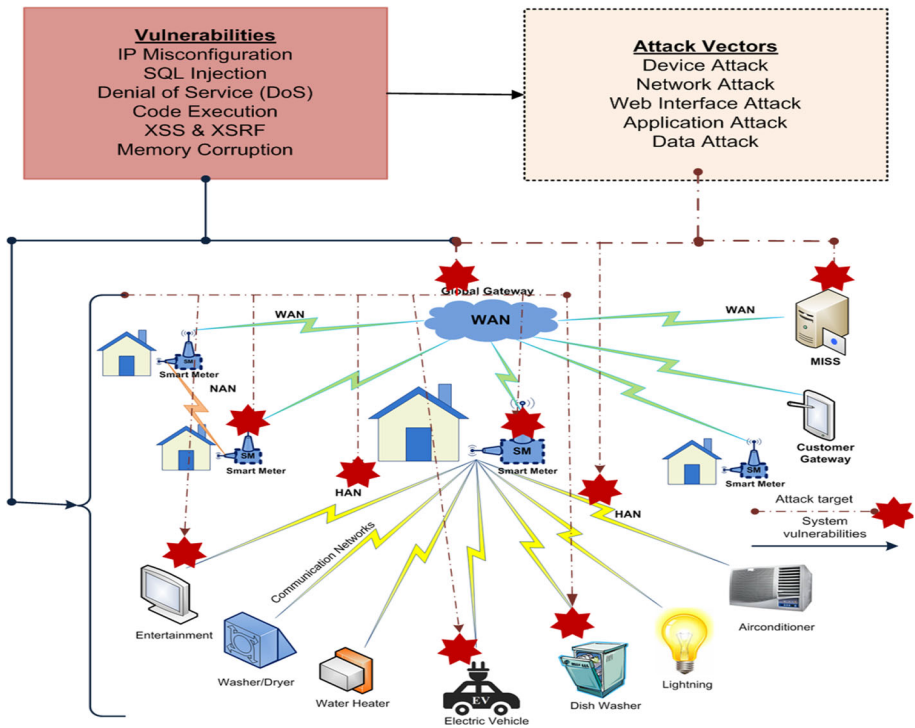


Fig. 8 Smart metering and corresponding threat attack [20]

Denial of Service Attack (Fig. 10)

DoS attack on the Application layer

Attack url:

<http://metering.smartmeter.com/metering/server/dashboard>

Tool: loadtest

(<https://www.npmjs.com/package/loadtest>) requires nodejs to be installed

Test parameter: \$ loadtest

<http://metering.aborsour.com/metering/server/dashboard> -t 50 -c 10 --rps 1000.

6 Discussion

The experiment above is based on a smart metering communication set-up as demonstrated in [20].

The idea of running both SQLi and DoS attacks on a smart metering system highlights their significant impact on distributed network system (see Figs. 9, 10). In the case of the SQL injection attack (experiment 1), a payload request was sent to the smart metering

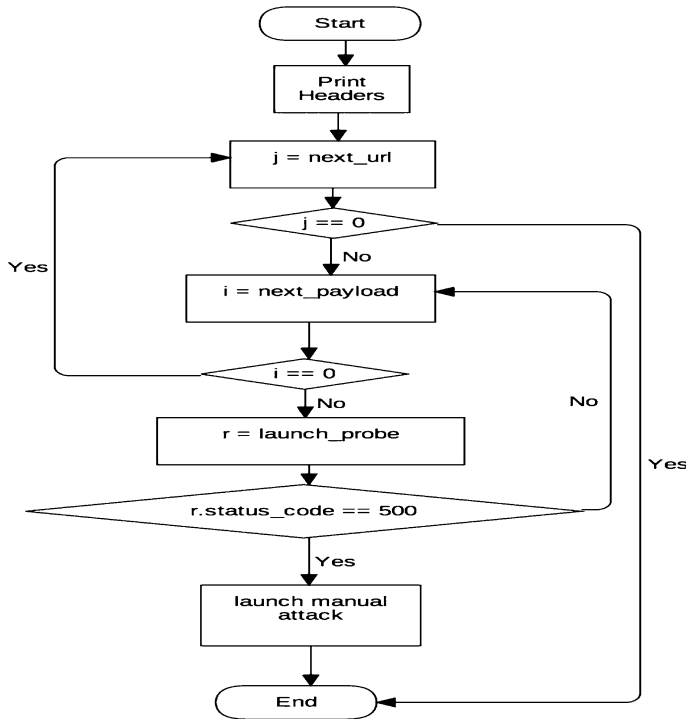


Fig. 9 Flowchart SQL injection attack

gateway to probe for vulnerabilities. The gateway acting as the main server in a neighbourhood network (NAN) setup, responded with an ACK message header paving way for a successful attack (compromised data) on the system. This means, SQL injection vulnerability in a smart metering communication architecture could allow remote authenticated users to execute arbitrary SQL commands via crafted serialized data both on the metering information system server (MISS) in both HAN and NAN smart metering setups (Fig. 9). For example, SQL injection vulnerability in the login page in a user interface device could allow remote attackers to execute arbitrary SQL commands via a crafted URL.

In the second experiments (DoS attack) (see Fig. 10), we executed multiple (abnormal) remote requests (1000) to a Metering Information System Server (MISS) for concurrent connections in 50 s. MISS is found in HANs and NANs of smart metering setup. Our result shows the server crushing resulted in executing some arbitrary codes (Fig. 10). A similar example is documented in [21]. In this attack, buffer overflow in the Point-to-Point Protocol over the Ethernet (PPPoE) module in the customer gateway when CHAP authentication is configured on the server, allow remote attackers to cause a denial of service or execute arbitrary code via crafted packets sent during authentication. For instance, in CVE-2016-8666 [21], an IP stack in the Linux kernel allows remote attackers to cause a denial of service (stack consumption and panic) or possibly have an unspecified impact by triggering use of GRO functions (gro-receive and gro-complete) path for packets with tunnel stacking.

```

[ ] loadtest http://metering.aborsour.com/metering/server/dashboard -t 50 -c 10 --rps 1000
[Fri Oct 28 2016 14:58:57 GMT+0000 (GMT)] INFO Requests: 0, requests per second: 0, mean latency: 0 ms
[Fri Oct 28 2016 14:59:02 GMT+0000 (GMT)] INFO Requests: 174, requests per second: 36, mean latency: 2431.2 ms
[Fri Oct 28 2016 14:59:02 GMT+0000 (GMT)] INFO Errors: 178, accumulated errors: 178, 100% of total requests
[Fri Oct 28 2016 14:59:07 GMT+0000 (GMT)] INFO Requests: 192, requests per second: 3, mean latency: 4185.1 ms
[Fri Oct 28 2016 14:59:07 GMT+0000 (GMT)] INFO Errors: 14, accumulated errors: 192, 100% of total requests
[Fri Oct 28 2016 14:59:12 GMT+0000 (GMT)] INFO Requests: 5162, requests per second: 994, mean latency: 1 ms
[Fri Oct 28 2016 14:59:12 GMT+0000 (GMT)] INFO Errors: 4970, accumulated errors: 5162, 100% of total requests
[Fri Oct 28 2016 14:59:17 GMT+0000 (GMT)] INFO Requests: 10159, requests per second: 1000, mean latency: 0.5 ms
[Fri Oct 28 2016 14:59:17 GMT+0000 (GMT)] INFO Errors: 4997, accumulated errors: 10159, 100% of total requests
[Fri Oct 28 2016 14:59:22 GMT+0000 (GMT)] INFO Requests: 15161, requests per second: 1001, mean latency: 0.5 ms
[Fri Oct 28 2016 14:59:22 GMT+0000 (GMT)] INFO Errors: 5002, accumulated errors: 15161, 100% of total requests
[Fri Oct 28 2016 14:59:27 GMT+0000 (GMT)] INFO Requests: 20162, requests per second: 1000, mean latency: 0.5 ms
[Fri Oct 28 2016 14:59:27 GMT+0000 (GMT)] INFO Errors: 5001, accumulated errors: 20162, 100% of total requests
[Fri Oct 28 2016 14:59:32 GMT+0000 (GMT)] INFO Requests: 25162, requests per second: 1000, mean latency: 133.5 ms
[Fri Oct 28 2016 14:59:32 GMT+0000 (GMT)] INFO Errors: 4991, accumulated errors: 25163, 100% of total requests
[Fri Oct 28 2016 14:59:37 GMT+0000 (GMT)] INFO Requests: 30161, requests per second: 1000, mean latency: 985.3 ms
[Fri Oct 28 2016 14:59:37 GMT+0000 (GMT)] INFO Errors: 4984, accumulated errors: 30137, 99.9% of total requests
[Fri Oct 28 2016 14:59:42 GMT+0000 (GMT)] INFO Requests: 34974, requests per second: 963, mean latency: 384.1 ms
[Fri Oct 28 2016 14:59:42 GMT+0000 (GMT)] INFO Errors: 4813, accumulated errors: 34950, 99.9% of total requests
[Fri Oct 28 2016 14:59:47 GMT+0000 (GMT)] INFO
[Fri Oct 28 2016 14:59:47 GMT+0000 (GMT)] INFO Target URL: http://metering.aborsour.com/metering/server/dashboard
[Fri Oct 28 2016 14:59:47 GMT+0000 (GMT)] INFO Max time (s): 50
[Fri Oct 28 2016 14:59:47 GMT+0000 (GMT)] INFO Concurrency level: 10
[Fri Oct 28 2016 14:59:47 GMT+0000 (GMT)] INFO Agent: none
[Fri Oct 28 2016 14:59:47 GMT+0000 (GMT)] INFO Requests per second: 1000
[Fri Oct 28 2016 14:59:47 GMT+0000 (GMT)] INFO
[Fri Oct 28 2016 14:59:47 GMT+0000 (GMT)] INFO Completed requests: 39870
[Fri Oct 28 2016 14:59:47 GMT+0000 (GMT)] INFO Total errors: 39846
[Fri Oct 28 2016 14:59:47 GMT+0000 (GMT)] INFO Total time: 50.001009381 s
[Fri Oct 28 2016 14:59:47 GMT+0000 (GMT)] INFO Requests per second: 797
[Fri Oct 28 2016 14:59:47 GMT+0000 (GMT)] INFO Mean latency: 259.4 ms
[Fri Oct 28 2016 14:59:47 GMT+0000 (GMT)] INFO
[Fri Oct 28 2016 14:59:47 GMT+0000 (GMT)] INFO Percentage of the requests served within a certain time
[Fri Oct 28 2016 14:59:47 GMT+0000 (GMT)] INFO 50% 1 ms
[Fri Oct 28 2016 14:59:47 GMT+0000 (GMT)] INFO 90% 271 ms
[Fri Oct 28 2016 14:59:47 GMT+0000 (GMT)] INFO 95% 285 ms
[Fri Oct 28 2016 14:59:47 GMT+0000 (GMT)] INFO 99% 4192 ms
[Fri Oct 28 2016 14:59:47 GMT+0000 (GMT)] INFO 100% 49209 ms (longest request)
[Fri Oct 28 2016 14:59:47 GMT+0000 (GMT)] INFO
[Fri Oct 28 2016 14:59:47 GMT+0000 (GMT)] INFO 100% 49209 ms (longest request)
[Fri Oct 28 2016 14:59:47 GMT+0000 (GMT)] INFO
[Fri Oct 28 2016 14:59:47 GMT+0000 (GMT)] INFO 500: 472 errors
[Fri Oct 28 2016 14:59:47 GMT+0000 (GMT)] INFO -1: 39374 errors
[Fri Oct 28 2016 14:59:47 GMT+0000 (GMT)] INFO Requests: 39870, requests per second: 975, mean latency: 571.4 ms
[Fri Oct 28 2016 14:59:47 GMT+0000 (GMT)] INFO Errors: 4896, accumulated errors: 39846, 99.9% of total requests

```

Fig. 10 Results (screenshot) of DoS attack

7 Conclusion

In this paper, an attempt has been made to evaluate the taxonomy of various system inherent vulnerabilities which expose IoT infrastructure and applications to various cyber threat vectors and make a case for research effort in this emerging technology. Our discussion involves the identification of different vulnerabilities inherent within IoT application and service domains. We executed two different attack scenarios (tests) on smart metering communication infrastructure setup. Our tests results show vulnerable IoT systems (being it application, hardware, software or firmware) could be abused by various threat actors via crafted vectors.

Finally, it is critical to continue the discussion while at the same time challenging device manufacturers and components' vendors to design, and implement solutions so as to counteract threats from cyber adversaries so as to guarantee consumer utmost trust in IoT innovation and transformation.

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